

reduce energy consumption based on building usage and occupancy. Because fixtures receive both data and power over a single Ethernet connection, this also eliminates the need for costly electrical wiring during construction or renovation," Joshi adds.

As connected LED lighting allows building managers to monitor and analyse building usage data in real time, they can identify efficiencies and control other facilities such as air-conditioning, heating and meeting room availability. "This can drive down energy consumption by up to 80 per cent compared to conventional lighting, allowing businesses to slash their carbon footprint and drastically reduce their running costs," informs Sumit.

As per a recent market report, almost 87 per cent of an average office worker's time is spent indoors. As people now spend less time outdoors, they consequently have less exposure to the natural rhythms of daylight. Connected lighting can enable offices to design indoor lighting to mimic the effects of natural daylight in an office, helping to optimise people's circadian rhythms, which can thereby improve their overall mood and productivity. If we consider the case of an R&D or a technology organisation in India, even a two per cent improvement in productivity can lead to an incremental increase of 50,000 rupees in revenue per employee every year, while reduced absenteeism and added health benefits can save around 15,000 rupees per employee every year.

Talking about the challenges, Anuj says, "It takes time for any new technology to penetrate the market. Customers should focus on the total cost of ownership. It should be noted that the new features go a long way in optimising energy usage and bring down the operating costs."

Case Study: Smart lighting benefits for Edge Building in Amsterdam

A good example of innovative smart LED lighting is the Edge Building in Amsterdam, Netherlands, which is an innovative, multi-tenant office building. The architect's goal was to create an intuitive, comfortable and productive environment for employees that would inspire sustain-

able building designs around the world. Opened in 2015, it received an outstanding score of 98.36 per cent—the highest ever—from BREEAM, the world's leading design and assessment method for sustainable buildings.

A key aspect of the building's sustainable design is a connected lighting system from Philips that enables employees to personalise the lighting and temperature at their workspaces via a smartphone app. The system also provides data on activities in the workspace, which further helps to reduce the CO₂ footprint.

The system uses nearly 6500 connected LED luminaires to create a 'digital ceiling' in the building's 15 storeys. With integrated sensors in 3000 of these luminaires that work with Philips Envision lighting management software, the system captures, stores, shares and distributes information throughout the illuminated space. Facility managers use the software to visualise and analyse this data, track energy consumption and streamline maintenance operations. The system uses 750 Power-over-Ethernet (PoE) switches to connect lighting fixtures to the building's IT network. With PoE, Ethernet cables send both power and data to the luminaires, eliminating the need for separate power cabling.

Towards a smart and connected office

Organisations exploring connected lighting systems must first decide the desired functionalities like scheduling, zoning, dimming and app-based control. The availability of wireless connected systems allows these benefits to be accessed for the LED retrofit market (conventional-to-LED upgrade, which is 83 per cent of the lighting market).

It is very difficult to put a specific number in terms of investment requirement. Typically, for MSMEs, there is 30-40 per cent incremental investment for a basic connected system from the basic LEDification. The ROI too depends upon the existing conventional system, utility charges and other factors. Typically, ROI takes around 2.5 to 3 years in the case of a basic connected system for MSMEs. **EFY**

—Nidhi Arora, executive editor, EFY

"The Whole (Education) Market Is Virtually Untapped In India"

► Success of any technology depends on its importance for the society at large. As India has the world's largest youth population, educational institutions play an important role in the national growth. Nidhi Arora, executive editor at EFY, in conversation with Reekrit Serai, co chair at Confederation of Indian Industry, explores technology adoption in the Indian education sector and possible business opportunities

Q. What are the popular technologies driving the current and future wave of education?

A. There are many trends picking up steam, but the major ones that will make any lasting impact are 3D printing (provided through Makerspaces or Atal Tinkering Labs in India), virtual reality (VR), augmented reality (AR), mixed reality (MR), Internet of Things (IoT), artificial intelligence (AI) including deep and machine learning, gamification and a gradual shift towards flip learning.

Q. How do these technologies benefit the education system and students at large?

A. In these disruptive times, children should get access to new technologies as their future employability and success depend on being able to use